

Vision-Based Localization for Micro-Continuum Surgical Robots: A Sim-to-Real Benchmark and Adaptation Study

Abstract—Micro-surgical robots enable dexterous manipulation in highly confined anatomical spaces and are increasingly explored for delicate procedures such as retinal membrane peeling. Their miniature continuum mechanisms make them promising platforms for partial surgical automation through small incisions. However, reliable automation requires precise localization of the robot tip relative to fragile tissue, yet motor encoders and kinematic models alone cannot achieve the required precision under compliance and hysteresis. Vision-based localization can provide robust external feedback, yet remains underexplored. We present a modular synthetic data generation toolkit that models the visual and physical complexity of a micro-surgical robotic environment. Leveraging this toolkit, we perform a unified benchmark and adaptation study for micro-continuum robots using paired synthetic and real data. Our simulation toolkit generates annotated RGB frames by projecting simulator (x, y, z) landmarks under controlled variation in retinal texture and illumination and manually annotated real videos. Using this dual-domain setup, we benchmark 13 localization methods spanning *classical*, *heatmap*, *regression*, and *detection paradigms* under sim-only transfer, real-only training, and sim-to-real finetuning. Results on a fixed held-out real test set show that resolution and pretraining are the dominant drivers of performance: Integral Regression improves from 40.7 px at 512×512 , and further to 21.5 px with sim-to-real finetuning; a multi-resolution ensemble reaches 16.3 px. These results establish a strong baseline and provide practical guidance for data-efficient deployment of vision-based localization in micro-continuum surgery.

I. INTRODUCTION

Micro-continuum surgical robots have emerged as a promising solution for highly confined anatomical workspaces [1], with applications spanning retinal micro-surgery [2], neurovascular intervention [3], and endoscopic surgery [4]. Despite their dexterity advantages, these systems share a fundamental control challenge: nonlinear hysteresis arising from tendon slack, internal friction, and material compliance complicates precise tip positioning [5].

Accurate tip localization is therefore essential for enabling automation and ensuring safe operation in microsurgical environments. Tasks such as membrane peeling or catheter navigation require precise knowledge of the instrument tip relative to delicate anatomical structures, where even sub-millimeter errors can cause tissue damage [6]. Reliable localization is thus a key requirement for both safe teleoperation and future autonomous surgical behaviors.

Vision-based localization offers an external feedback mechanism that bypasses explicit hysteresis modeling by operating directly on observed motion outcomes. However, deploying such methods in intraocular surgical settings remains fundamentally constrained by the scarcity of labeled data.

The purpose of this work is to evaluate the efficacy of the state-of-the-art vision-based localization methods for micro-continuum robots. On the data side, acquiring synchronized RGB images with ground-truth keypoint annotations from micro-continuum surgical robots under real intraocular procedures is impractical at scale. As a solution, simulation can help mitigate data shortages by providing synthetic data. To this end, prior work introduced a physics based toolkit for synthetic data generation. On the domain adaptation side, prior work demonstrates that the sim-to-real gap in lighting, tissue appearance, and depth availability substantially degrades localization performance on real microscope imagery [7] — yet no systematic study exists on how different model families respond to this gap, or how efficiently real annotations finetuning can close it.

In this work, we address these challenges through three contributions spanning dataset construction, systematic evaluation, and domain adaptation for vision-based localization of micro-continuum surgical robots:

- **Synthetic and real dataset construction:** An automated pipeline built on the AMBF intraocular environment [7] generates annotated RGB frames for vision-based localization. 2D keypoints are derived from simulator (x, y, z) states through calibrated camera projection. The synthetic dataset spans 12 controlled visual conditions, combining 4 retinal background textures with 3 lighting configurations. In addition, we introduce a manually annotated real intraocular RGB dataset of a micro-continuum surgical robot operating inside real and phantom porcine eyes under a surgical microscope.
- **Multi-method benchmark:** A systematic evaluation of 13 vision-based localization methods spanning classical, heatmap, regression, and detection paradigms, benchmarked on both simulated and real intraocular data under zero-shot sim transfer and real-only training regimes, establishing the first comprehensive performance baseline for this domain
- **Domain adaptation study:** A sim-to-real finetuning analysis across model families and numbers of manually annotated real images, demonstrating that synthetic pre-training provides a complementary axis of improvement beyond the saturation point of real data alone, with practical guidelines for data-efficient deployment

II. RELATED WORK

A. Micro-Continuum Surgical Robots

Micro-continuum robots have emerged as a compelling solution for surgical tasks requiring access to confined

anatomical workspaces with high dexterity. Unlike rigid-link manipulators, continuum robots achieve motion through elastic deformation or tendon-driven actuation, enabling enhanced dexterity and compact profiles suitable for minimally invasive procedures [5].

In retinal and intraocular surgery, the confined geometry of the eye and the fragility of retinal tissue place extreme demands on positioning accuracy. Systems such as the Base Robot (BR) [8] address tremor suppression for rigid instruments, while cable-driven continuum designs extend dexterity to access intraocular structures through small incisions [2]. Concentric tube robots represent another prominent continuum paradigm, with applications spanning minimally invasive surgical manipulation and deep-access interventions [1], sharing the fundamental challenges of hysteresis and compliance present in cable-driven designs. In endoscopic [9] and catheter-based intervention, similar continuum architectures have been explored extensively, with hysteresis compensation remaining an open control problem across all these systems [3].

A common thread across all these platforms is that kinematic modeling alone is insufficient for precise tip localization under hysteresis [10], motivating external sensing and vision-based feedback as a complementary approach [5]. Nevertheless, vision-based localization in these settings is complicated by the confined, texturally complex workspaces in addition to the lack of large-scale annotated datasets, yet these challenges are directly addressed by our benchmark.

B. Simulation Frameworks for Surgical Robotics

Physics-based simulation has become an essential tool for surgical robotics research, enabling scalable data generation, policy training, and system evaluation without the cost and risk of real hardware operation. Early simulation environments such as the dVRK simulator [11] focused primarily on replicating the kinematics of commercial surgical platforms for teleoperation research.

More recent frameworks have emphasized learning-oriented design. SurRoL [12] provides a reinforcement learning-compatible environment for the da Vinci Research Kit with task-relevant surgical subtasks. ORBIT-Surgical [13] extends this paradigm using GPU-parallelized simulation to support both reinforcement and imitation learning at scale for augmented surgical dexterity. FF-SRL [14] demonstrates that GPU-accelerated deformable tissue simulation can reduce RL training time by an order of magnitude compared to CPU-based alternatives.

For vision-oriented data generation specifically, the Asynchronous Multi-Body Framework (AMBF) [15] provides real-time physics simulation with native support for synchronized RGB-D capture, segmentation, and pose annotation, making it particularly suitable for intraocular robotic scenarios [7]. Building on AMBF, prior work developed a modular synthetic data generation toolkit for the localization of a custom-built micro-continuum surgical robot (CMSR), demonstrating sub-millimeter pose tracking with simulated depth and zero-shot sim-to-real detection [7]. However, that

work evaluates only two vision methods and does not study the effect of incorporating real data. Our work extends this line by providing the first systematic benchmark of vision-based localization methods across both simulated and real intraocular data. In addition, we characterize how synthetic pretraining interacts with limited quantities of manually annotated real images.

C. Localization Methods

Vision-based localization for surgical robots spans a broad range of sensing modalities and learning paradigms. We group the 13 evaluated image-based methods into the following categories:

- **Classical baselines** are low-capacity, non-deep-learning models including Template Matching [16], and Color Segmentation [17].
- **Heatmap-based methods** are models that predict dense keypoint likelihood maps followed by soft-argmax coordinate extraction, including SimpleBaseline [18], HR-Net [19], Stacked Hourglass [20], CenterNet [21], and U-Net [22].
- **Direct regression methods** are models that regress coordinates directly from global image features, including ResNet Direct [23] and ViT Direct [24].
- **Integral regression methods** are hybrid approaches that bridge heatmap prediction and coordinate regression through differentiable expectation operators, including Integral Regression [25], ViTPose [26], and DINOv2 Regression [27].
- **Detection-based localization** is a transformer-style detection framework applied to keypoint localization, represented by DETR [28].

III. METHODOLOGY

A. Simulation Toolkit Design

The simulation environment is built on the Asynchronous Multi-Body Framework (AMBF) [7], which provides real-time physics simulation with tendon-driven actuation, synchronized RGB-D capture, and native support for external control interfaces. The environment comprises three modular components — the robot model, the eyeball model, and the virtual camera/microscope — each defined by AMBF Description Format (ADF) files that specify geometry, inertial properties, and joint structure independently, allowing substitution of any component without modifying the others.

Robot model. The eye snake is modeled as a chain of 11 disc-like segments connected by alternating revolute joints, replicating the physical cable-driven continuum structure. A two fingers grip is attached: one fixed and one movable, consistent with the real CMSR hardware. Geometry and inertial parameters are matched to the physical robot’s dimensions and fabricated segment masses, ensuring that the simulated kinematics reproduce the real system’s bending behavior under actuation.

Eyeball model. A 12 mm-radius spherical eyeball is constructed in Blender¹ and exported to ADF format. The inner

¹<https://www.blender.org/>

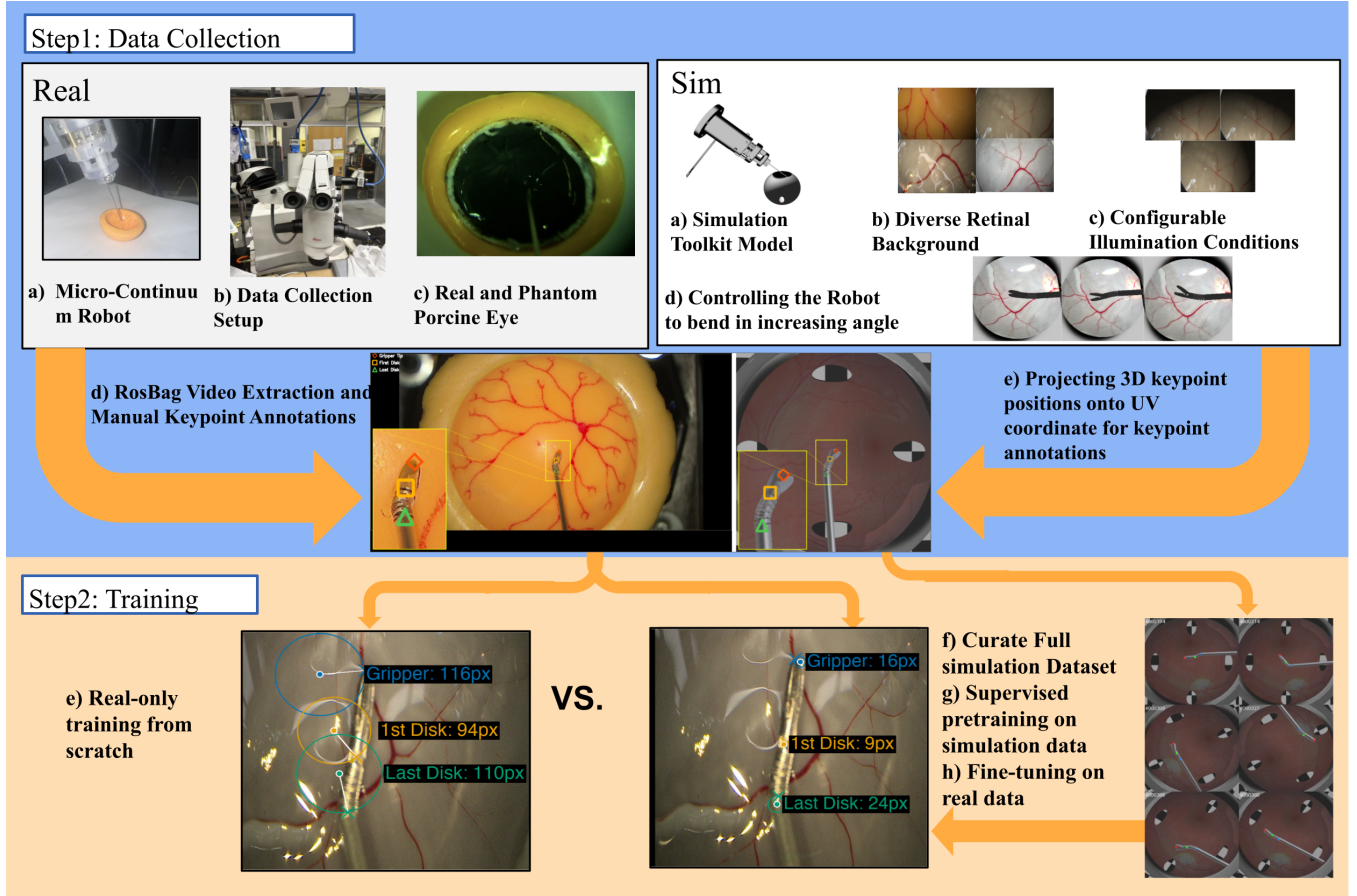


Fig. 1. Overview of the dual-domain data collection and training pipeline. **Step 1 – Data Collection.** *Real* (left): the micro-continuum surgical robot is mounted on a BR base under a Leica surgical microscope and operated inside real and phantom porcine eyes; RGB video is extracted from ROS bags and annotated with three manual keypoints. *Simulation* (right): an AMBF-based toolkit models the robot geometry with configurable retinal background textures and illumination conditions; the robot is programmatically controlled through increasing bending angles, and 3D keypoint positions are projected onto UV image coordinates for automatic annotation. **Step 2 – Training.** Real-only training from scratch (left) is compared against sim-to-real finetuning (right), where models are first pretrained on the curated simulation dataset (f–g) then finetuned on real data (h), yielding substantially lower localization error across all three keypoints.

surface accepts interchangeable retinal texture maps derived from real porcine intraocular images, providing anatomically grounded background appearance while allowing controlled texture variation across data collection runs.

Camera model. A virtual camera is mounted above the eyeball to replicate the viewpoint of a Leica surgical microscope (Leica Microsystems, Germany). Intrinsic parameters — focal length, principal point, and lens distortion coefficients — are obtained via standard checkerboard calibration of the physical microscope and transferred directly to the virtual camera, ensuring that simulated images share the same projection geometry as real captures. A distortion plugin applies the calibrated radial and tangential distortion coefficients at render time.

Robot control. Rather than modeling the full multi-stage hysteresis of the physical system, the control layer employs a simplified constant-curvature kinematic formulation. The entire bending section is approximated as a circular arc parameterized by total pitch and yaw angles ($\phi_{\text{total}}, \psi_{\text{total}}$),

with curvature vector:

$$\vec{\kappa} = \left[\frac{\phi_{\text{total}}}{L}, \frac{\psi_{\text{total}}}{L}, 0 \right]^T, \quad \theta = \|\vec{\kappa}\|L \quad (1)$$

where L is the arc length. The resulting tip pose in the world frame is computed via the Rodriguez rotation formula [7] and sent to the simulation. Bending angles are constrained to $(-20^\circ, 20^\circ)$ in both pitch and yaw to match the physical operating envelope. Forward and inverse kinematics with a remote-center-of-motion (RCM) constraint are implemented in Python, enabling efficient posture sampling for large-scale data generation without per-step manual control.

B. Simulation Data collection

Using the toolkit described above, we generate a dataset by sampling 300 robot configurations within the reachable workspace and rendering each configuration under controlled visual variations.

Visual diversity is introduced along two controlled axes. Lighting is varied across three discrete configurations spanning front illumination to backlighting, modulating overall

illumination levels and shadow distributions. Four distinct retinal background textures derived from real porcine eye images are applied to the eyeball inner surface, reducing overfitting to a single anatomical appearance. The combination yields 12 distinct visual configurations per robot pose, as illustrated in Fig. 1.

For each rendered frame, the 2D image-space coordinates (u, v) of each keypoint are obtained by projecting the simulator’s ground-truth 3D keypoint positions $\mathbf{p}_{\text{target}}^{\text{world}}$ into the camera frame via:

$$(u, v)^T = \mathbf{K} \cdot \mathbf{P} \cdot (\mathbf{T}_{\text{camera}}^{\text{world}})^{-1} \cdot \mathbf{p}_{\text{target}}^{\text{world}} \quad (2)$$

where $\mathbf{K}$ is the camera intrinsic matrix, $\mathbf{P}$ is the projection matrix, and $\mathbf{T}_{\text{camera}}^{\text{world}}$ is the camera pose in world coordinates [7]. Labels are generated for three keypoints — gripper tip, first disk (segment 11), and last disk (segment 1) — matching the keypoint definitions used in real data collection to enable direct cross-domain comparison without annotation format discrepancy.

C. Real Data Collection

Real data is collected using a micro-continuum surgical robot mounted on a BR base [8], which provides stable positioning of the overall surgical apparatus. The continuum segment is inserted into a physical eyeball model through a small incision, replicating the intraocular workspace of a clinical retinal procedure. Enucleated porcine eyes are used as the anatomical substrate, providing natural retinal texture, intraocular fluid, and tissue reflectance properties that are not fully replicated in simulation.

A Leica surgical microscope is positioned above the eye and provides the imaging viewpoint for all real data collection, consistent with the camera model used in simulation [7]. RGB video is recorded directly from the microscope output at 1920×1080 resolution at a fixed focal plane. The robot is manually actuated through a range of pitch and yaw configurations across recording sessions to capture diverse bending states and visual conditions. No depth sensor is used, as accurate depth acquisition under surgical microscope optics at millimeter scales remains an open hardware challenge [7]; ground-truth labels are therefore restricted to 2D image-space keypoint coordinates.

Frames are extracted from each video sequence and annotated using the same three-keypoint protocol as simulation, labeling the gripper tip, first disk (segment 11), and last disk (segment 1) when visible. Frames where any keypoint is occluded or outside the field of view are excluded. All annotations are performed manually using a custom labeling interface and reviewed for consistency.

D. Dataset Statistics and Splits

Fig. 2 summarizes both real and simulation data domains. The simulation dataset comprises 300 robot configurations rendered across 12 visual conditions (4 retinal textures $\times$ 3 lighting setups), yielding 300 annotated frames with automatically projected 2D gripper labels. These are expanded to 900

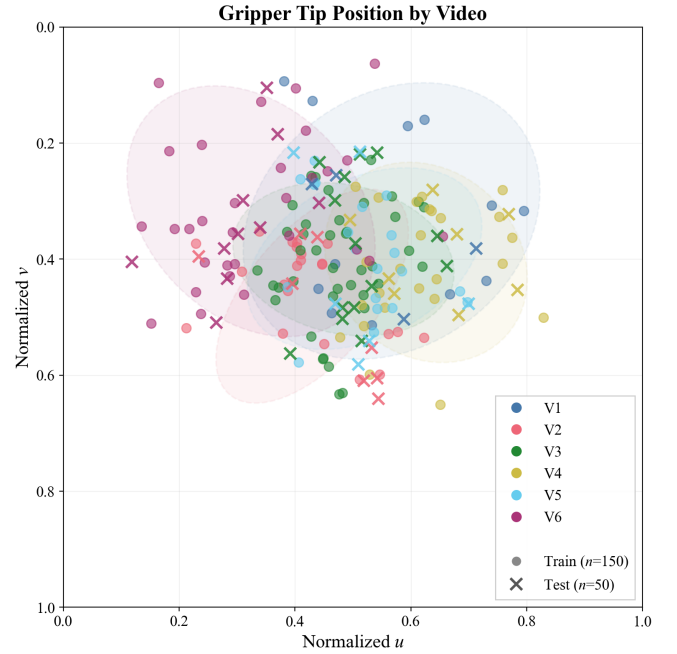


Fig. 2. Gripper tip positions across 150 training (circles) and 50 held-out test (crosses) images. V1–V6 denote six recording sessions with different camera placements and robot poses. Ellipses show per-video 1σ covariance. The train/test split preserves spatial coverage across all sessions.

samples via domain randomization augmentations (brightness reduction, Gaussian blur, scaling, noise injection).

The real dataset comprises 6 video sequences (V1–V6) recorded during ex-vivo porcine eye experiments with the CMSR, totaling 400 extracted frames. Of these, 217 frames contain complete 3-keypoint annotations (grripper tip, first disk, last disk) verified as visible and unoccluded; the remaining 183 frames are unlabeled or partially annotated. Fig. 2 illustrates both the spatial distribution of gripper tip positions across videos and the partitioning pipeline.

From the 217 annotated frames, we construct a fixed held-out test set of 50 frames and a training pool of up to 150 frames, with stratified sampling to maintain proportional video representation in both splits. The 17 remaining annotated frames are held in reserve. From the training pool, nested subsets of $N \in \{30, 60, 90, 120, 150\}$ frames are constructed for data efficiency analysis, with strict inclusion ($N = 30 \subset N = 60 \subset \dots \subset N = 150$) enforced at each scale to ensure that improvements from additional data are not confounded by changes in sample composition. In all experiments, models are evaluated exclusively on the 50-frame real test set (1920×1080 native resolution), providing a consistent cross-domain evaluation benchmark regardless of training domain or resolution.

E. Training and Fine-tuning Strategy

We evaluate each approach under two training regimes: *Real-Only*, trained from scratch on N real images; and *Sim→Real Fine-tuning (FT)*, pretrained on 300 simulation images then fine-tuned on N real images. All models use

AdamW [29] with cosine annealing, $\text{lr} = 10^{-3}$, and 150 epochs unless otherwise noted.

Initialization strategy. A critical finding is that naively loading all simulation-pretrained weights for fine-tuning can *degrade* performance below the real-only baseline when real data is scarce. We trace this to a backbone corruption effect: simulation pretraining overwrites general-purpose ImageNet [30] features with simulation-specific representations that cannot be recovered from ≤ 150 real samples.

This motivates a backbone-dependent initialization strategy:

- *Approaches with ImageNet-pretrained backbones* (Integral Regression, SimpleBaseline, ResNet Direct, CenterNet, DETR, Image UV+3D): we apply **hybrid initialization**—retain the original ImageNet backbone weights and load only the sim-pretrained task heads (deconvolution layers, regression heads, or decoders). This preserves the general visual features while benefiting from sim-learned task structure.
- *Approaches without pretrained backbones* (Hourglass, HRNet, U-Net, ViT Direct, ViTPose): since no general-purpose features exist to preserve, we load all simulation-pretrained weights and fine-tune the entire network.
- *DINOv2 Regression*: the self-supervised DINOv2 backbone is kept frozen throughout both pretraining and fine-tuning; only the lightweight regression head is updated.

Multi-resolution ensemble. To exploit the complementary strengths of different input resolutions, we train Integral Regression [25] models independently at 512×512 , 768×768 , and 1024×1024 pixels. At inference, each model runs at 5 scales (training resolution ± 32 and ± 64 px), and all $K = 3 \times 5 = 15$ predictions are averaged in normalized $[0, 1]$ coordinate space:

$$(\hat{u}, \hat{v}) = \frac{1}{15} \sum_{k=1}^3 \sum_{s \in \mathcal{S}_k} f_k^{(s)}(I) \quad (3)$$

where $f_k^{(s)}$ denotes model k evaluated at scale s .

IV. EXPERIMENTAL EVALUATION

A. Training Protocol

We study two input resolutions:

224 px. Models train for 30 epochs with batch size 8. Real-only training uses $\text{lr} = 10^{-3}$; finetuning uses $\text{lr} = 10^{-4}$ (lower rate to preserve pretrained features). Data efficiency is evaluated over nested subsets $N \in \{30, 60, 90, 120, 150\}$ (Table I).

512 px. Models train for 200 epochs with batch size 4, $\text{lr} = 10^{-4}$, and a 5-epoch linear warmup. The longer schedule compensates for the smaller effective batch at higher resolution. All models use $N = 150$ real images (Table II). We additionally evaluate a *multi-resolution ensemble* that averages normalized predictions from Integral Regression models trained independently at 512, 768, and 1024 px.

Unified protocol. To ensure fair comparison across architectures, all deep learning approaches share the same

hyperparameters within each resolution-regime setting. We deliberately avoid per-model tuning: our goal is to evaluate architectural inductive biases under controlled conditions, not optimized configurations. This may disadvantage architectures that require specialized schedules (e.g., vision transformers typically benefit from longer warmup and lower learning rates). The two classical baselines (Template Matching and Color Segmentation) use their own fitting procedures and are unaffected by these hyperparameters.

B. Metrics

Mean pixel error (px). We report the Euclidean distance between predicted and ground-truth keypoints in pixel space, $e = \sqrt{(\hat{u} - u)^2 + (\hat{v} - v)^2}$, where (u, v) are pixel coordinates on a 1920×1080 image. Models predict normalized coordinates in $[0, 1]$, which are scaled back by image dimensions for error computation. Results are reported per keypoint (gripper, first_disk, last_disk) and overall (mean across all three keypoints over all test samples).

Why pixel error (not mm). Real data provides only 2D image-space labels, so pixel error is the most direct and consistent metric. Because models predict in normalized $[0, 1]$ space, pixel error is resolution-independent. For reference, at a typical working distance of a CMSR, $1 \text{ px} \approx 0.015 \text{ mm}$ (so $21.5 \text{ px} \approx 0.32 \text{ mm}$).

Test protocol. We evaluate on 50 held-out real frames from 6 video sequences at 1920×1080 resolution. There is zero overlap between train and test frames (verified by frame index). Each model is evaluated on the same fixed test set across all experiments, and all reported numbers are from this test set (not validation).

““

C. Results and Analysis

Tables I and II present the full benchmark. We organize our analysis into five findings ordered by practical significance.

Finding 1: Sim-to-real finetuning is effective at 512 px but unreliable at 224 px. At 512 px (Table II), 6 of 9 deep learning approaches improve with sim pretraining and finetuning. Integral Regression shows the largest gain: $40.7 \rightarrow 21.5 \text{ px}$ (-47%), followed by DETR from 95.9 to 76.9 px (-20%) and Hourglass from 70.2 to 60.0 px (-15%). Three approaches show slight degradation: CenterNet ($+5\%$), DINOv2 ($+3\%$), and U-Net ($+1\%$)—all approaches that either already achieve low real-only error or use a frozen pretrained backbone.

At 224 px with $N = 150$ (Table I), the pattern reverses: 6 of 11 approaches *worsen* with finetuning. The top-performing approaches degrade most—Hourglass rises from 62.3 to 74.2 px ($+19\%$) and ResNet Direct from 68.2 to 82.3 px ($+21\%$). Only approaches with high real-only error benefit: DETR improves from 212.8 to 112.8 px (-47%) and SimpleBaseline from 182.6 to 120.0 px (-34%). At low resolution, the information bottleneck limits how well simulation features can transfer, causing negative transfer for approaches that already fit the real distribution.

TABLE I

DATA EFFICIENCY AT 224×224 PX: MEAN PIXEL ERROR (PX) ON 50 HELD-OUT REAL TEST IMAGES (1920×1080). REAL-ONLY: TRAINED FROM SCRATCH ON N REAL IMAGES ($LR=10^{-3}$). FT: SIM-PRETRAINED (300 SIM IMAGES) THEN FINETUNED ON N REAL IMAGES ($LR=10^{-4}$). ALL RUNS USE 30 EPOCHS, BATCH SIZE 8, COSINE SCHEDULER. BEST RESULT PER APPROACH IN **BOLD**.

Approach	Category	Backbone	Params	Real-Only (224px)					Sim→Real Finetuning (224px)				
				$N=30$	$N=60$	$N=90$	$N=120$	$N=150$	$N=30$	$N=60$	$N=90$	$N=120$	$N=150$
Hourglass	Heatmap	Hourglass-2	6.6M	210.7	226.3	153.4	81.5	62.3	210.0	128.3	108.0	95.7	74.2
ResNet Direct	Regression	ResNet-50	24.3M	147.3	248.4	137.0	73.5	68.2	197.9	161.2	118.7	81.9	82.3
CenterNet	Heatmap	ResNet-18	14.0M	190.9	139.5	109.5	136.0	78.9	152.6	120.9	99.9	92.8	81.7
Integral Regression	Regression	ResNet-50	34.0M	166.3	130.7	266.5	153.7	89.6	248.4	206.1	107.3	91.5	93.4
SimpleBaseline	Heatmap	ResNet-50	34.0M	208.4	585.1	199.2	101.3	182.6	207.6	186.8	155.7	127.7	120.0
DETR	Detection	ResNet-50	12.2M	213.1	152.4	205.7	115.3	212.8	206.5	164.6	135.2	107.7	112.8
U-Net	Heatmap	U-Net	31.0M	232.4	214.9	196.2	203.1	203.4	219.6	202.1	186.7	176.1	189.9
ViTPose	Regression	ViT-Small	17.2M	220.6	209.3	188.1	186.5	186.8	240.9	222.7	194.8	188.6	172.1
HRNet	Heatmap	HRNet-W32	28.5M	229.4	231.3	202.1	193.6	182.4	231.6	210.5	210.5	191.6	193.9
DINOv2 Regression	Regression	DINOv2-S	13.1M	212.1	204.8	214.3	183.3	212.5	231.1	220.5	207.6	201.6	186.8
ViT Direct	Regression	ViT-Small	11.2M	269.1	212.9	201.5	190.8	189.4	261.1	252.2	212.2	192.5	212.6
Color Segmentation	Classical	—	163	694.4	821.7	396.1	581.4	305.7	312.4	300.5	289.2	281.7	293.9
Template Matching	Classical	—	4K	443.0	251.7	282.8	242.8	230.5	—	—	—	—	—

TABLE II

HIGH-RESOLUTION RESULTS AT 512×512 PX WITH $N=150$ REAL TRAINING IMAGES. MEAN 2D PIXEL ERROR (PX) ON 50 HELD-OUT REAL TEST FRAMES AT 1920×1080 . *Real-Only*: TRAINED FROM SCRATCH ON REAL DATA. *Sim→Real FT*: PRETRAINED ON 300 SIMULATION IMAGES THEN FINETUNED ON REAL DATA. $\Delta\%$: RELATIVE CHANGE IN ERROR FROM FINETUNING (NEGATIVE = IMPROVED, POSITIVE = DEGRADED). BEST RESULT PER APPROACH IN **BOLD**.

Approach	Real-Only	Sim→Real FT	$\Delta\%$
Integral Reg.	40.7	21.5	−47
DINOv2 Reg.	40.7	41.8	+3
Hourglass	70.2	60.0	−15
SimpleBaseline	83.3	72.5	−13
DETR	95.9	76.9	−20
ResNet Direct	90.5	79.6	−12
CenterNet	80.6	84.7	+5
U-Net	83.8	84.4	+1
HRNet	152.8	133.4	−13
ViT Direct	N/A ^b	N/A ^b	—
ViTPose	N/A ^b	N/A ^b	—
Template Match.	358.0	—	—
Color Seg.	841.0	—	—
<i>Multi-res. ensemble (IntReg FT):</i> 512+768+1024			—
	—	16.3	—

^bViT positional embeddings incompatible at 512px.

Finding 2: Finetuning corrects harder samples & rescues challenging cases Per-sample analysis reveals that sim-to-real preferentially fixes the samples where real-only training fails most (Fig. 3). For Integral Regression at 512 px, finetuning **improves 46 of 50** test samples; the hardest quartile (by real-only error) reduces error by 16.5px on average. The correlation between real-only difficulty and finetuning improvement is positive for 6 of 7 models. Analysis of ground-truth keypoint positions shows that hard samples exhibit $3\times$ higher positional variance and $1.8\times$ larger keypoint spread (225 vs. 127 px)—these are extended or unusual poses

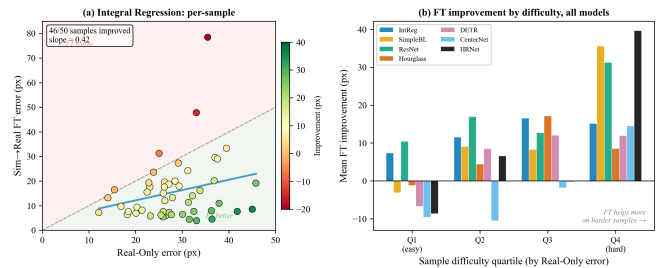


Fig. 3. Per-sample analysis of sim-to-real finetuning benefit at 512 px. (a) Scatter of Real-Only vs. Sim→Real FT error for Integral Regression across all 50 test frames. Points below the dashed diagonal ($y=x$) indicate FT improvement; color encodes absolute improvement in px. **46 of 50 samples improve with finetuning** (slope = 0.42), with larger gains on samples where real-only error is highest. (b) Mean FT improvement per difficulty quartile (Q1 = easiest, Q4 = hardest by real-only error) across 7 models. Finetuning consistently degrades or provides negligible benefit on easy samples (Q1) while strongly helping hard samples (Q4), with improvements from 40 px for HRNet and 32 px for ResNet Direct. This pattern indicates that simulation data compensates for the limited pose coverage of real training data, specifically rescuing underrepresented configurations.

that the 150 real training images undersample. Simulation data, which covers comprehensive elevation angles with uniform pose sampling, provides exactly this missing coverage, explaining why pretraining on simulation specifically rescues the challenging cases.

Finding 3: Data scaling at 224px is non-monotonic.

Table I reveals that most approaches do not improve monotonically with more training data at 224 px. SimpleBaseline varies from 208 px ($N=30$) to 585 px ($N=60$, catastrophic overfitting) to 101 px ($N=120$, best) to 183 px ($N=150$). Integral Regression spikes to 267 px at $N=90$ despite achieving 131 px at $N=60$. Only 2 of 11 deep learning approaches (CenterNet, Hourglass) scale approximately monotonically. Sim pretraining stabilizes this behavior: CenterNet FT decreases monotonically from 153 px ($N=30$) to 82 px ($N=150$), and DETR FT from 207 to 113 px. This suggests simulation pretraining regularizes training even when it does not reduce the final error.

Finding 4: Resolution provides the largest single im-

provement. Comparing best results across resolutions, upgrading from 224 to 512 px yields larger gains than any architecture change at fixed resolution. Integral Regression improves from 89.6 to 40.7 px (−55%), DINOv2 from 183.3 to 40.7 px (−78%), and U-Net from 186.7 to 83.8 px (−55%). At 224 px, the gap between the best deep learning approach (Hourglass, 62.3 px) and worst (ViT Direct, 189.4 px) is 3.0×; at 512 px with finetuning, the gap between best (Integral Regression, 21.5 px) and worst (HRNet, 133.4 px) widens to 6.2×—resolution amplifies rather than levels architectural differences.

Finding 5: Coordinate decoding determines finetuning benefit. Among heatmap approaches at 512 px, the decoding strategy predicts finetuning gain. Soft-argmax (Integral Regression) computes a differentiable weighted average over the heatmap, enabling end-to-end gradient flow and the largest finetuning improvement (−47%). Grid-quantized argmax approaches (CenterNet) snap predictions to discrete grid locations, disrupting gradient flow from sim-pretrained features and showing no benefit (+5%). Multi-scale deconvolution approaches (SimpleBaseline, Hourglass) achieve moderate finetuning gains (−13% to −15%). Direct regression methods (DINOv2) bypass spatial decoding entirely: DINOv2 matches Integral Regression at 40.7 px real-only but gains nothing from finetuning (+3%), suggesting its frozen backbone already captures sufficient features without adaptation.

Finding 6: Multi-resolution ensemble provides complementary gains. Averaging predictions from three Integral Regression models trained at 512, 768, and 1024 px—each with five-scale test-time augmentation (15 total scales)—achieves 16.3 px, a 24% reduction over the best single model (21.5 px). Different resolutions favor different keypoints: 1024 px excels for the small gripper tip, while 512 px better captures the larger disk structures. This ensemble requires no additional training, only inference-time computation.

V. CONCLUSION

We presented a systematic benchmark of 13 vision-based approaches across five categories—heatmap regression, direct regression, detection, segmentation, and classical methods—for localizing three keypoints on the CMSR from monocular RGB images. Using clean, non-overlapping train/test splits with 30–150 real annotated frames and 300 simulation images, we identified three actionable findings.

First, input resolution is the single most impactful factor: upgrading from 224×224 to 512×512 px reduces Integral Regression error from 89.6 px to 40.7 px (55%), a larger improvement than any architecture change or data scaling at 224 px. Seven of thirteen approaches improve at higher resolution, with gains strongest for models with large pre-trained backbones (DINOv2 80%, U-Net 57%). Conversely, classical methods collapse—Template Matching prediction spread drops to 4% of ground-truth variation, effectively outputting a fixed point regardless of input.

Second, simulation pretraining and resolution interact multiplicatively: at 512 px, sim-to-real finetuning reduces

Integral Regression error a further 47% (40.7→21.5 px), whereas finetuning provides no consistent benefit at 224 px. The combination of high resolution and pretraining yields our best single-model result of 21.5 px (0.32 mm at typical working distance).

Third, multi-resolution ensembling provides complementary gains: averaging predictions from three Integral Regression models trained at 512, 768, and 1024 px with five-scale test-time augmentation each achieves 16.3 px overall error, a 24% reduction over the best single model, as different resolutions capture complementary spatial information (1024 px excels for the small gripper tip; 512 px for the larger disk keypoints).

These results establish a practical recipe for surgical robot keypoint localization with limited annotations: train at the highest feasible resolution using soft-argmax heatmap regression, leverage simulation pretraining for finetuning, and ensemble across resolutions when latency permits. Future work will extend evaluation to in-vivo sequences, investigate temporal consistency across video frames, and explore resolution-adaptive architectures that avoid the fixed positional embedding limitations of current ViT-based methods.

REFERENCES

- [1] P. E. Dupont, J. Lock, B. Itkowitz, and E. Butler, “Design and control of concentric-tube robots,” *IEEE Transactions on Robotics*, vol. 26, no. 2, pp. 209–225, 2010.
- [2] Anonymous, “Title and authors removed for the 2026 iros double-anonymous review process,” 2020.
- [3] J. Burgner-Kahrs, D. C. Rucker, and H. Choset, “Continuum robots for medical applications: A survey,” *IEEE Transactions on Robotics*, vol. 31, no. 6, pp. 1261–1280, 2015.
- [4] H.-S. Yoon, H.-J. Cha, J. Chung, and B.-J. Yi, “Compact design of a dual master-slave system for maxillary sinus surgery,” in *2013 IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2013, pp. 5027–5032.
- [5] R. J. I. Webster and B. A. Jones, “Design and kinematic modeling of constant curvature continuum robots: A review,” *The International Journal of Robotics Research*, vol. 29, no. 13, pp. 1661–1683, 2010.
- [6] M. Balicki, A. Uneri, I. Iordachita, J. Handa, P. Gehlbach, and R. Taylor, “Micro-force sensing in robot assisted membrane peeling for vitreoretinal surgery,” in *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2010*, T. Jiang, N. Navab, J. P. W. Pluim, and M. A. Viergever, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2010, pp. 303–310.
- [7] Anonymous, “Title and authors removed for the 2026 iros double-anonymous review process,” 2026.
- [8] A. Uneri, M. A. Balicki, J. Handa, P. Gehlbach, R. H. Taylor, and I. Iordachita, “New steady-hand eye robot with micro-force sensing for vitreoretinal surgery,” in *2010 3rd IEEE RAS & EMBS International Conference on Biomedical Robotics and Biomechanics (BioRob)*. Tokyo, Japan: IEEE, 2010, pp. 814–819.
- [9] B. P. M. Yeung and T. Gourlay, “A technical review of flexible endoscopic multitasking platforms,” *International Journal of Surgery*, vol. 10, no. 7, pp. 345–354, 2012. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1743919112001069>
- [10] B. Y. Cho, D. S. Esser, J. Thompson, B. Thach, R. J. Webster, and A. Kuntz, “Accounting for hysteresis in the forward kinematics of nonlinearly-routed tendon-driven continuum robots via a learned deep decoder network,” *IEEE Robotics and Automation Letters*, vol. 9, no. 11, pp. 9263–9270, 2024.
- [11] Z. Chen, A. Deguet, S. Vozar, A. Munawar, G. Fischer, and P. Kazanzides, “Interfacing the da vinci research kit (dvrk) with the robot operating system (ros),” in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA) Workshops*, Seattle, WA, USA, 2015.

- [12] J. Xu, B. Li, B. Lu, Y.-H. Liu, Q. Dou, and P.-A. Heng, "Surrol: An open-source reinforcement learning centered and dvrk compatible platform for surgical robot learning," in *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021, pp. 1821–1828.
- [13] Q. Yu, M. Moghani, K. Dharmarajan, V. Schorp, W. C.-H. Panitch, J. Liu, K. Hari, H. Huang, M. Mittal, K. Goldberg, and A. Garg, "Orbit-surgical: An open-simulation framework for learning surgical augmented dexterity," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 15 509–15 516.
- [14] D. Dall'Alba, M. Naskret, S. Kamińska, and P. Korzeniowski, "Ff-srl: High performance gpu-based surgical simulation for robot learning," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 8378–8384.
- [15] A. Munawar and G. S. Fischer, "An asynchronous multi-body simulation framework for real-time dynamics, haptics and learning with application to surgical robots," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2019, pp. 6268–6275.
- [16] R. Brunelli, *Template matching techniques in computer vision: theory and practice*. John Wiley & Sons, 2009.
- [17] H.-D. Cheng, X. H. Jiang, Y. Sun, and J. Wang, "Color image segmentation: advances and prospects," *Pattern recognition*, vol. 34, no. 12, pp. 2259–2281, 2001.
- [18] B. Xiao, H. Wu, and Y. Wei, "Simple baselines for human pose estimation and tracking," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018.
- [19] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 5686–5696.
- [20] A. Newell, K. Yang, and J. Deng, "Stacked hourglass networks for human pose estimation," in *European Conference on Computer Vision (ECCV)*. Springer, 2016, pp. 483–499.
- [21] X. Zhou, D. Wang, and P. Krähenbühl, "Objects as points," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.
- [22] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015.
- [23] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
- [24] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations (ICLR)*, 2021.
- [25] X. Sun, B. Xiao, F. Wei, S. Liang, and Y. Wei, "Integral human pose regression," in *Proceedings of the European conference on computer vision (ECCV)*, 2018, pp. 529–545.
- [26] Y. Xu, J. Zhang, Q. Zhang, and D. Tao, "Vitpose: Simple vision transformer baselines for human pose estimation," *Advances in neural information processing systems*, vol. 35, pp. 38 571–38 584, 2022.
- [27] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby *et al.*, "Dinov2: Learning robust visual features without supervision," *arXiv preprint arXiv:2304.07193*, 2023.
- [28] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2020.
- [29] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *ICLR*, 2019.
- [30] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "Imagenet: A large-scale hierarchical image database," in *2009 IEEE conference on computer vision and pattern recognition*. Ieee, 2009, pp. 248–255.